Practical No. 7

Aim: Programs on Services, Notification and Broadcast Receivers.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout android:layout_height="match_parent"
   android:layout_width="match_parent"
   xmlns:android="http://schemas.android.com/apk/res/android">

      <Button
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:id="@+id/btnClick"
       android:text="Click Me"
       android:layout_centerInParent="true"
       android:textSize="25dp"/>

</RelativeLayout>
```

activity_main2.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout android:layout_height="match_parent"
   android:layout_width="match_parent"
   xmlns:android="http://schemas.android.com/apk/res/android" >

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:id="@+id/tv"
      android:text="Notification Page"
      android:textSize="25dp"
      android:layout_centerInParent="true"/>

</RelativeLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />
    <application
       android:allowBackup="true"
       android:dataExtractionRules="@xml/data_extraction_rules"
       android:fullBackupContent="@xml/backup_rules"
       android:icon="@mipmap/ic_launcher"
       android:label="@string/app_name"
       android:roundIcon="@mipmap/ic_launcher_round"
```

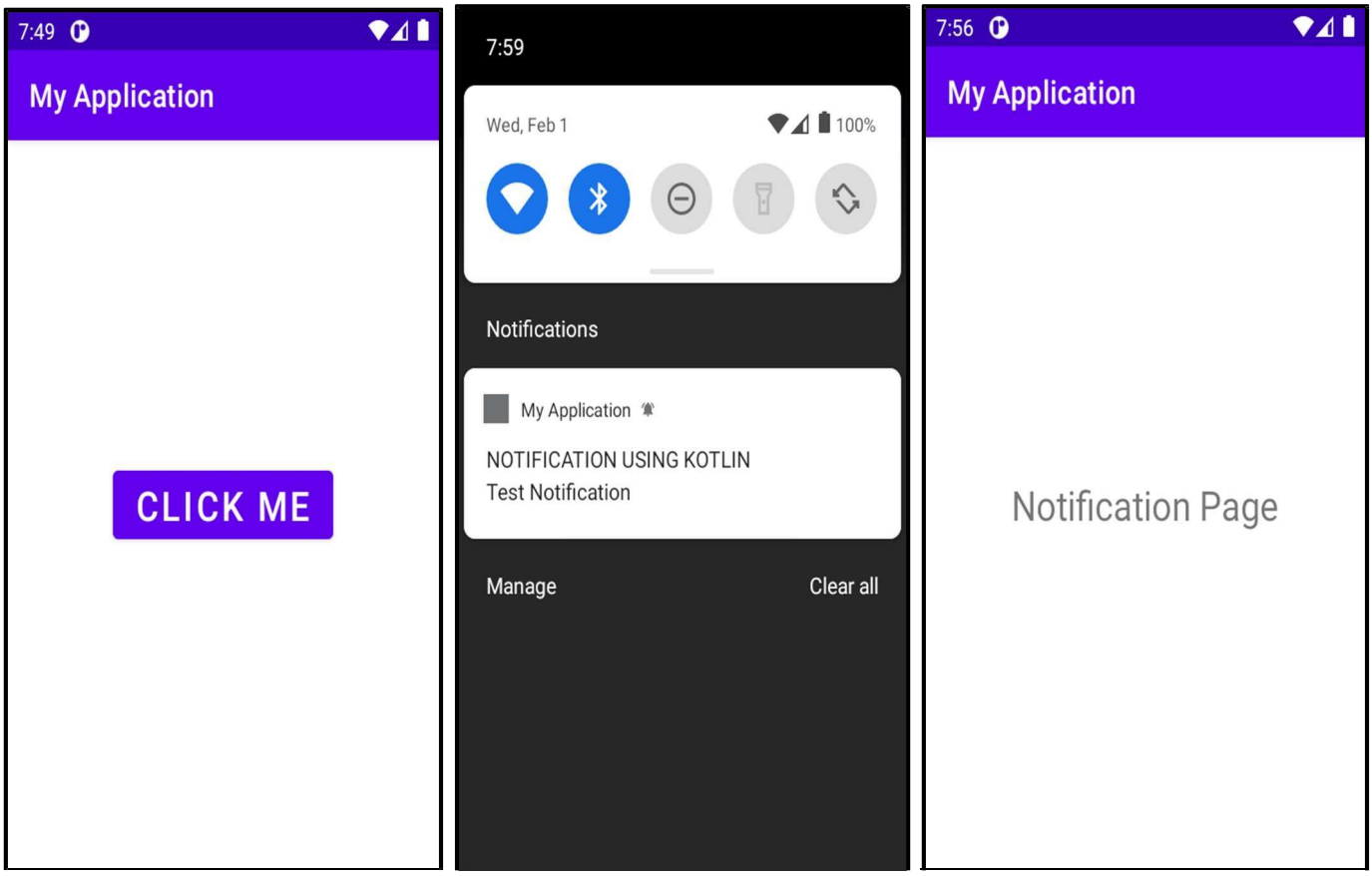
```
        android:supportsRtl="true"
        android:theme="@style/Theme.Final2">
    <activity
            android:name=".MainActivity2"
            android:exported="false" />
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>

 </manifest>
```

MainActivity.kt

```
package com.example.myapplication

import android.annotation.SuppressLint
import android.app.Notification
import android.app.NotificationChannel
import android.app.NotificationManager
import android.app.PendingIntent
import android.content.Context
import android.content.Intent
import android.graphics.Color
import android.os.Build
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button

class MainActivity : AppCompatActivity() {

    lateinit var btnClick: Button
    lateinit var notificationManager: NotificationManager
    lateinit var notificationChannel: NotificationChannel
    lateinit var builder: Notification.Builder
    private val channelId = "12345"
    private val description = "Test Notification"

    @SuppressLint("MissingInflatedId")
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        btnClick = findViewById(R.id.btnClick)
        notificationManager = getSystemService(Context.NOTIFICATION_SERVICE) as
NotificationManager
```

```
        btnClick.setOnClickListener {
            val intent = Intent(this, MainActivity2::class.java)
            val pendingIntent = PendingIntent.getActivity(this, 0, intent,
PendingIntent.FLAG_IMMUTABLE)

            if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
                notificationChannel = NotificationChannel(channelId, description,
NotificationManager.IMPORTANCE_HIGH)
                notificationChannel.lightColor = Color.BLUE
                notificationChannel.enableVibration(true)
                notificationManager.createNotificationChannel(notificationChannel)
                builder = Notification.Builder(this, channelId)
                    .setContentTitle("NOTIFICATION USING KOTLIN")
                    .setContentText("Test Notification")
                    .setSmallIcon(R.drawable.ic_launcher_background)
                    .setContentIntent(pendingIntent)
            }
            notificationManager.notify(1234, builder.build())
        }
    }
}
```

Output


7:49
My Application
CLICK ME
7:59
Wed, Feb 1
100%
Notifications
My Application
NOTIFICATION USING KOTLIN
Test Notification
Manage
Clear all
7:56
My Application
Notification Page